Homework 1 – One Sample Inference

Human toddlers, starting at around 18 months old, can learn novel object labels passively by overhearing third-party interactions without being directly addressed. For example, if one adult says "hand me an apple" and another does so, the child might learn to identify an apple and associate it with the word "apple". This process is driven by developing sociocognitive capacities such as gaze following, conversation monitoring, intention understanding, and perspective-taking. Of course, this process is more efficient when adults use simplified sentences, emphasis, enthusiasm, and other non-verbal emphasis. Dogs share some human-like social and cognitive abilities they learn being immersed in our households. For example, many family dogs learn actions like "sit." Dror et al. (2026) explored whether dogs could be capable of passive label learning, similar to toddlers. Specifically, they were interested in Gifted Word Learner (GWL) dogs that can learn the names of hundreds of objects and perform at a word-learning level comparable to 18 months olds. To learn whether GWL dogs learn object labels by passively overhearing interactions between humans the researchers performed an experiment where = 10 dogs were exposed to a condition where two family members sitting across from each other passing the toy back and forth. The caretaker addressed the second person using simple sentence frames and non-verbal cues, neither directly to the dog. This exposure occurred daily for eleven days with two new toys.

- Toy A: Days 1-3; 9
- Toy B: Days 5-7; 10
- Rest: Day 4, 8, 11

On the twelfth day, out of the caretaker's sight, the two new toys were placed alongside nine familiar toys. The caretaker requested the toys verbally twenty-four times – twelve times for the two new toys, and twelve times for the nine familiar toys.

1 Question 1

1.1 Part a

The researchers are asking two questions: (1) whether performance in the overhearing condition is above 50% chance, and (2) whether the difference between the direct and overhearing conditions differs from zero.

Before choosing a test we summarize each data set.

```
overhear <- c(1.0000, 1.0000, 0.7500, 1.0000, 0.4167,
              0.8333, 0.7500, 0.8333, 0.8333, 1.0000)

pdiff <- c(-0.0833, 0.0000, -0.0833, -0.0833, 0.5833,
           0.0833, -0.2500,  0.1667, 0.0000, -0.4167)

summarize.data <- function(x){
  tibble(
    n                = length(x),
    Mean             = mean(x),
    SD               = sd(x),
    Median           = median(x),
    IQR              = IQR(x),
    Skewness         = e1071::skewness(x),
    `Excess Kurtosis` = e1071::kurtosis(x),
    Min              = min(x),
    Max              = max(x)
```

```
  )
}
```

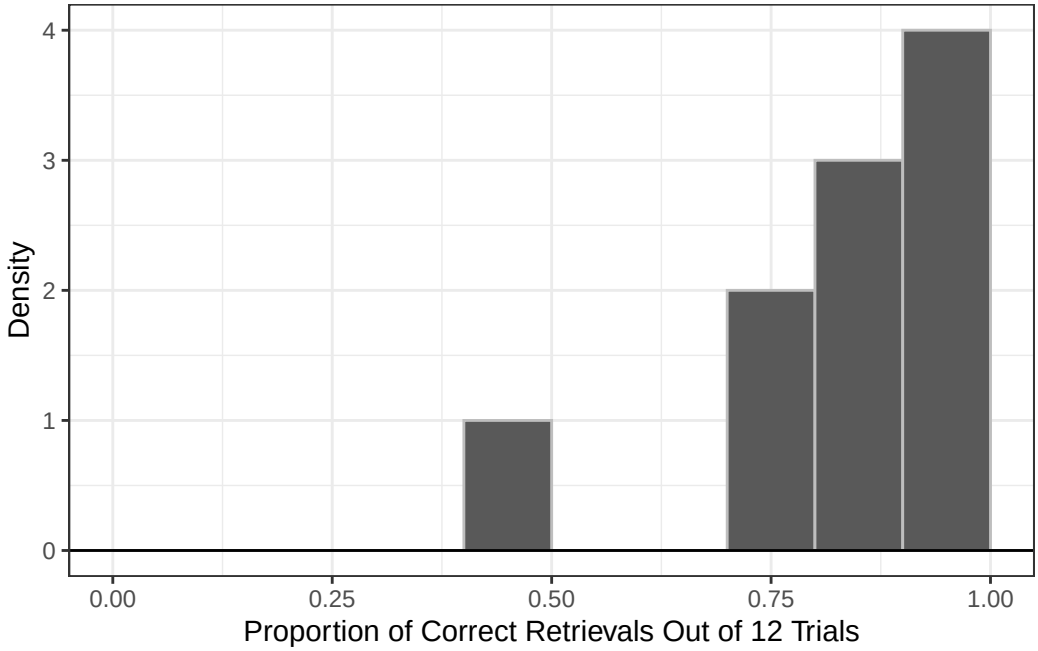
```
summarize.data(overhear) |> knitr::kable(digits = 2)
```

Table 1: Descriptive statistics for the overhearing condition

n	Mean	SD	Median	IQR	Skewness	Excess Kurtosis	Min	Max
10	0.84	0.18	0.83	0.23	-1.04	0.22	0.42	1

```
ggplot(data = tibble(overhear), aes(x = overhear)) +
  geom_histogram(aes(y = after_stat(density)),
                 breaks = seq(0, 1, 0.1),
                 color = "grey") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Proportion of Correct Retrievals Out of 12 Trials",
       y = "Density")
```

Figure 1: Proportion of correct retrievals in the overhearing condition




```
summarize.data(pdiff) |> knitr::kable(digits = 2)
```

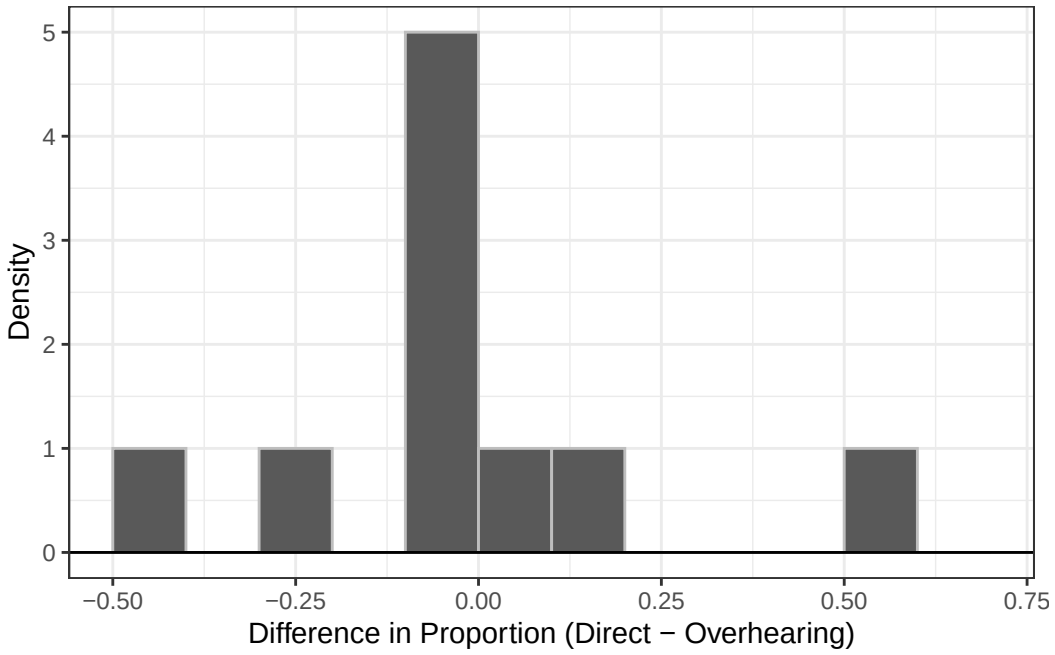
Table 2: Descriptive statistics for the paired differences

n	Mean	SD	Median	IQR	Skewness	Excess Kurtosis	Min	Max
10	-0.01	0.26	-0.04	0.15	0.7	0.15	-0.42	0.58

```
ggplot(data = tibble(pdiff), aes(x = pdiff)) +
  geom_histogram(aes(y = after_stat(density)),
                 breaks = seq(-0.5, 0.7, 0.1),
```

```
                 color = "grey") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Difference in Proportion (Direct - Overhearing)",
       y = "Density")
```

Figure 2: Paired differences (direct − overhearing)



1.1.1 The overhearing data

We cannot use the t-test because the distribution is left skewed and bounded, and the sample size is too small to invoke the Central Limit Theorem.

We cannot use the exact Wilcoxon test because there are ties and the distribution is skewed. We also cannot use the approximate Wilcoxon test because the sample size is less than 50. So, the authors did not make a good choice in selecting Wilcoxon test and without specifying exact or approximate.

We cannot use the z-test because $np_0 = 10(0.50) = 5$, below the required 15.

I would use the exact Binomial test, which involves transforming the data into Bernoulli data (i.e., whether each dog retrieved the correct new toys in more than half of the 12 trials).

1.1.2 The paired differences data

We cannot use the t-test because the distribution is skewed and the sample size is too small for the Central Limit Theorem.

We cannot use the Wilcoxon test because there are zeros and ties, and the overall distribution is not symmetric.

We cannot use the exact Binomial test here because we are interested in how large the difference between conditions is, and reducing the data to signs treats a difference of 0.0833 and a difference of 0.5833 the same, losing important information.

I would use a randomization test. This uses the actual magnitudes rather than just the signs, and makes no assumption about the shape of the population distribution.

1.2 Part b

We want to know whether the dogs retrieved the new toys above 50% chance, and whether there is a difference between the two conditions: direct address and overhearing.

1.2.1 Research Question 1: Is performance above chance?

Transform the data into Bernoulli form, recording whether each dog retrieved the new toys correctly on more than half of its twelve trials, and use the exact Binomial test.

Conditions. The variable is categorical after transformation. The dogs come from separate households, so observations are independent across dogs. The sample is a good representation of the Gifted Word Learner dogs. The exact Binomial test imposes no condition on sample size.

- H_0: $p = 0.50$
- H_a: $p \neq 0.50$

```
overhear.bernoulli <- as.numeric(overhear > 0.50)

rq1.result <- binom.test(x = sum(overhear.bernoulli),
                         n = length(overhear.bernoulli),
                         p = 0.50,
                         alternative = "two.sided",
                         conf.level = 0.95)

rq1.result$p.value
```

```
[1] 0.02148438
```

```
rq1.result$conf.int
```

```
[1] 0.5549839 0.9974714
attr(,"conf.level")
[1] 0.95
```

```
(cohens.h <- 2*asin(sqrt(0.9)) - 2*asin(sqrt(0.5)))
```

```
[1] 0.9272952
```

Of the ten dogs tested, nine performed above chance across their twelve trials. This proportion was statistically discernible from the chance level of 0.5, $p = .021$, 95% CI [.555, .997]. The effect size was large, Cohen's $h = 0.927$.

There is statistically discernible support for the alternative hypothesis. The dogs retrieved the new toys above chance in the overhearing condition, which indicates that they can associate labels with objects from overheard speech alone. The interval, .555 to .997, is wide. The dogs clearly did better than chance, but with only ten of them we cannot tell whether they are a little better or almost perfect.

1.2.2 Research Question 2: Do the two conditions differ?

Use a randomization test, which uses the observed magnitudes and makes no assumption about the shape of the population distribution.

Conditions. The variable is quantitative. The dogs are independent of one another. Under the null hypothesis the two conditions are interchangeable within a dog, so we can flip the sign,

- H_0: $\mu_X = 0$
- H_a: $\mu_X \neq 0$

```
set.seed(7272)
R <- 10000
```

```r

null.dist <- rep(NA, R)
for(i in 1:R){
  signs <- sample(x = c(-1, 1), size = length(pdiff), replace = TRUE)
  null.dist[i] <- mean(signs * pdiff)
}

(observed <- mean(pdiff))
```

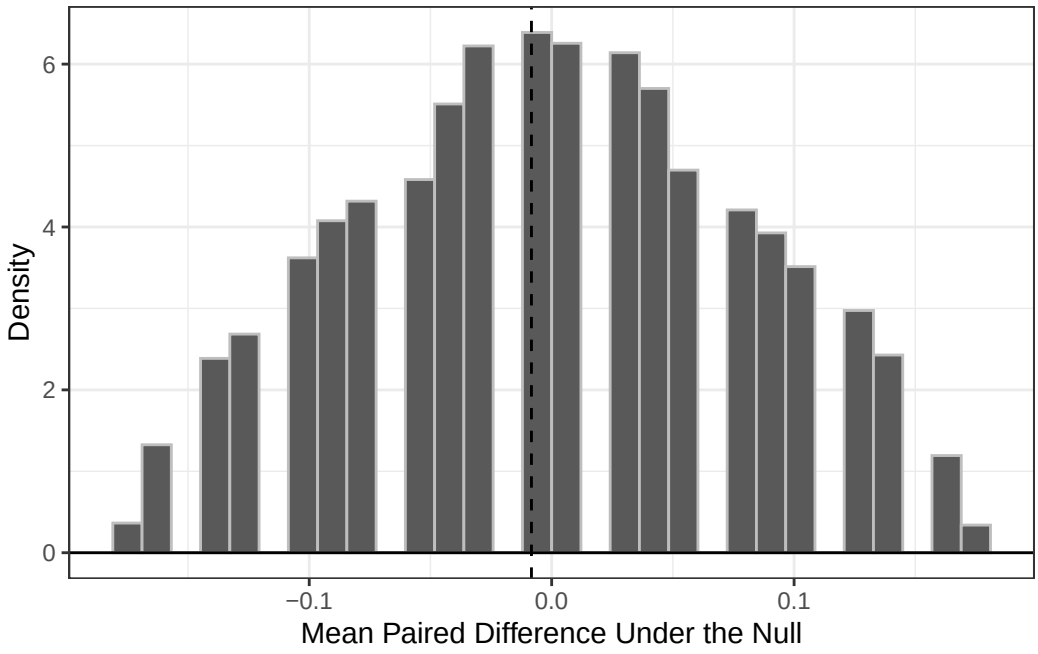
```
[1] -0.00833
```

```r
(rq2.pvalue <- mean(abs(null.dist) >= abs(observed)))
```

```
[1] 0.9847
```

```r
ggplot(data = tibble(null.dist), aes(x = null.dist)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 30, color = "grey") +
  geom_vline(xintercept = observed, linetype = "dashed") +
  geom_hline(yintercept = 0) +
  theme_bw() +
  labs(x = "Mean Paired Difference Under the Null",
       y = "Density")
```

Figure 3: Null distribution of the mean paired difference, with the observed value marked



A confidence interval for the mean difference is obtained by bootstrapping.

```r
set.seed(7272)
boot.means <- rep(NA, R)
for(i in 1:R){
  curr.sample <- sample(x = pdiff, size = length(pdiff), replace = TRUE)
  boot.means[i] <- mean(curr.sample)
}

(rq2.ci <- quantile(boot.means, probs = c(0.025, 0.975)))
```

```
    2.5%    97.5%
-0.15001  0.15833
```

```r
(hedges.g <- effectsize::hedges_g(x = pdiff, mu = 0))
```

```
Hedges' g |        95% CI
-------------------------
-0.03     | [-0.59, 0.54]
```

The mean paired difference observed in our sample was not statistically discernible from zero, $M = -0.008$, $SD = 0.265$, $p = .99$, 95% CI $[-0.15, 0.16]$. The effect size was negligible, Hedges' $g = -0.03$.

We do not have statistically discernible support for the alternative hypothesis. The dogs did about as well whether the caretaker spoke to them directly or they only overheard the labels.

2 Question 2

The researchers evaluated two new-toy trials per dog across = 10 dogs. In the overheard condition, the dogs retrieved all the new toys correctly on the first try. That is, the proportion of times the dogs retrieved the new toys they had only learned through overhearing was 1.0.

2.1 Part a

The binomial model is a sum of Bernoulli trials. Thus, each X_i must have a binary outcome, the number of trials n must be fixed, the trials must be independent, and the probability of the event of interest p must be the same in every trial.

First, retrieval is binary: the dog either brings the requested toy or it does not. Second, the design fixed $n = 10$ dogs with two trials each, so n was fixed. Third, I think the trails are not independent. A dog that learned the label should succeed on both trials, and a dog that did not should tend to fail on both. So the two outsomes seems to be linked in someways rahter than fully independent. Finally, I doubt the probability stays the same in every trial. Once the first toy has been retrieved, the total number of toy decrease by one. So next time, the dog is selecting among 10 toys. The probability will change.

2.2 Part b

Under author assumption that a single retrieval succeeds by chance with probability 0.5, and two retrievals are independent,

$$P(X_i = 1) = 0.5 \times 0.5 = 0.25.$$

$$B = \sum_{i=1}^{10} X_i \sim \text{Binomial}(n = 10,\ p = 0.25),$$

```r
dbinom(x = 10, size = 10, prob = 0.25)
```

```
[1] 9.536743e-07
```

2.3 Part c

```r
n <- 10
p <- 0.25
mu <- n * p
sigma <- sqrt(n * p * (1 - p))

approx <- pnorm(q = 10.5, mean = mu, sd = sigma) -
          pnorm(q = 9.5,  mean = mu, sd = sigma)
exact  <- dbinom(x = 10, size = 10, prob = 0.25)

c(exact = exact, approx = approx, ratio = exact / approx)
```

```
       exact       approx        ratio
9.536743e-07 1.567447e-07 6.084254e+00
```

The approximation is 1.567×10^{-7}, and the exact value from Part b is 9.537×10^{-7}. For abs error, they differ by about 8×10^{-7}, which is pretty small. While for rel error, the approximation is poor. The approximation understates the true value by roughly 84%.

2.4 Part d

When all trials are successes, $\hat{p} = 1$ and the standard error $\sqrt{\hat{p}(1-\hat{p})/n}$ becomes zero, so the z interval collapses to the single point $[1, 1]$. The sample size condition fails too, since $n(1-\hat{p}) = 0$. When these conditions are not met we can use the exact Binomial distribution.

$$H_0 : p = 0.25 \qquad H_a : p > 0.25$$

```r
test.result = binom.test(x = 10,
                         n = 10,
                         p = 0.25,
                         alternative = "greater",
                         conf.level = 0.95)
test.result$p.value
```

```
[1] 9.536743e-07
```

```r
test.result$conf.int
```

```
[1] 0.7411344 1.0000000
attr(,"conf.level")
[1] 0.95
```

There is statistically discernible support for the alternative hypothesis: the dogs performed better than chance.

3 References

Dror, Shany, Ádám Miklósi, Boglárka Morvai, Andreea-Silvia Năstase, and Claudia Fugazza. 2026. "Dogs with a Large Vocabulary of Object Labels Learn New Labels by Overhearing Like 1.5-Year-Old Infants." *Science* 391 (6781): 160–63.